Project: Worked examples
Title: Accumulator sizing (gas law)
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1. Scope and method

The pre-charge gas volume V0 of a hydraulic accumulator that delivers, or absorbs, a liquid volume ΔV between the lowest working pressure p1 and the highest p2, from $p \cdot V^n$ = const with absolute pressures and an exponent n between 1 (isothermal) and 1.4 (adiabatic), the gas either charged as fast as it is discharged or (a held accumulator, the default for an emergency stroke) charged slowly and isothermally to p2 and then discharged at n; the volume scaled to the lowest operating temperature (T_max/T_min) and the pre-charge pressure at the charging temperature; the next size from your own size list; and the bar·litre product PS·V of the vessel's own maximum allowable pressure and volume against the PSSR Schedule 1 250 bar·L exception. Three modes: energy storage, shock and pulsation (screening only) and emergency stroke (ΔV is the stroke volume, linkable from the cylinder circuit). Not a pressure-vessel code check.

WHAT IT CALCULATES	RELATION
Absolute pressures	$p_a = p_{gauge} + p_{atm}$ (or as entered, in Absolute mode)
Gas volume, one exponent both ways (charged as fast as it is discharged)	V0 = ΔV / [(p0/p1)^(1/n) − (p0/p2)^(1/n)], n = 1 isothermal … 1.4 adiabatic
Gas volume, slow charge and discharge at n (a held accumulator)	charged isothermally to p2, V2 = V0·p0/p2, then discharged at n to p1, V1 = V2·(p2/p1)^(1/n), ΔV = V1 − V2, so V0 = ΔV·(p2/p0) / [(p2/p1)^(1/n) − 1]
Either volume × the real-gas factor f_{rg} (user input, at least 1) and × the cold-gas factor, because p0 is defined at T_{max} and the gas delivers less when cold	$(T_{max} + 273.15)/(T_{min} + 273.15)$ for the held accumulator, that ratio to the power 1/n for one exponent both ways
Pre-charge rules (records, applied to absolute pressures)	$p0 \le r_0 \cdot p1$ at T_{max} (default 0.9, bladder type) and $p2/p0(T_{min}) \le r_p$ (default 4), $p0(T_{min}) = p0 \cdot (T_{min} + 273.15)/(T_{max} + 273.15)$
Pre-charge at the charging temperature	$p0(T_c) = p0(T_{max}) \cdot (T_c + 273.15)/(T_{max} + 273.15)$
Next size	the smallest listed size not below V0
PSSR bar-litre product	PS·V, PS = the vessel's maximum allowable pressure (bar gauge, entered, not below p2 gauge), V = the vessel volume entered (default the selected size); compared with 250 bar·L (PSSR Schedule 1, indication only; not available without PS and a volume)

Basis: 250 bar·litre exception, indication only[1](Schedule 1); Pressure equipment; not checked here[2]; Gas-charged accumulator sizing, $p \cdot V^n$ = const[3](Ch. on accumulators).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mode	Duty	Energy storage		assumption: Slow clamp-and-hold cycle: stored-energy duty.	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
dV	Liquid volume ΔV delivered, absorbed or (emergency stroke) needed for the stroke	14	L	spec: Hydraulic specification HS-12: clamp stroke plus leakage make-up, 14 L per cycle	0.001 … 1000 L
process	Gas process (By duty: an emergency stroke is charged slowly to p2 and discharged at n, the other duties use one exponent both ways)	By duty		default	any
p_{basis}	Pressures entered as	Absolute		assumption: Pressures entered as absolute bar, as the source numbers are quoted.	any
p_0	Gas pre-charge pressure p0, at the maximum operating temperature	90	bar a	spec: Hydraulic specification HS-12: pre-charge 90 bar at the maximum oil temperature	0 … 700 bar a
p_1	Lowest working pressure p1	100	bar a	spec: Hydraulic specification HS-12: lowest clamp pressure 100 bar	0 … 700 bar a
p_2	Highest working pressure p2	210	bar a	spec: Hydraulic specification HS-12: pump cut-out 210 bar	0 … 700 bar a
p_{atm}	Atmospheric pressure (gauge to absolute)	1.01325	bar a	assumption: Standard atmosphere, derived constant.	0.5 … 1.1 bar a
n	Gas exponent n of the discharge (1 isothermal, slow; 1.4 adiabatic, fast); with the same exponent both ways it is also the charge exponent	1		assumption: Isothermal: the clamp holds for minutes between cycles.	1 … 1.4

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
f_{rg}	Real-gas factor on V0 (1 = ideal gas; at least 1). It is the ratio of the real to the ideal gas volume for THIS pressure window, not the compressibility factor Z at p_2: from a gas-property source, $f_{rg} = Z_0 \cdot (1/p_1 - 1/p_2)/(Z_1/p_1 - Z_2/p_2)$ for an isothermal discharge, Z at p_0, p_1 and p_2, which is larger than Z_2 for a narrow window	1		assumption: Ideal gas; the highest pressure is just above 200 bar, so the sheet warns.	1 … 3
T_{max}	Maximum operating temperature (p0 applies here)	50	°C	spec: Hydraulic specification HS-12: oil temperature limit 50 °C	-40 … 150 °C
T_{min}	Minimum operating temperature (the gas is coldest and delivers least here; V0 is scaled by T_{max}/T_{min})	30	°C	assumption: Oil temperature range assumed 30 to 50 °C in service (invented for this example); the gas is coldest at 30 °C.	-40 … 150 °C
T_c	Charging temperature	20	°C	assumption: Charged in the workshop at 20 °C.	-40 … 100 °C
r_0	Pre-charge rule: p0 (at T_{max}) at most this fraction of p1, on absolute pressures (state whether the manufacturer's ratio is on gauge or absolute pressures)	0.9		reference: record: accumulator pre-charge rules (bladder type), to be taken from the manufacturer's guide	0.1 … 1
r_p	Pre-charge rule: p2/p0 (p0 at T_{min}) at most this ratio, on absolute pressures	4		reference: record: accumulator pre-charge rules (bladder type), to be taken from the manufacturer's guide	1 … 20
sizes	Available accumulator sizes (from the manufacturer's range)	4 row(s)		reference: record: illustrative size list invented for this example, to be replaced by the manufacturer's range	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
PS	Vessel maximum allowable pressure PS, bar gauge (nameplate or relief setting; not below p2 gauge; needed for the PSSR indication)	230	bar g	spec: Hydraulic specification HS-12: accumulator relief setting 230 bar gauge (vessel PS; illustrative, to be replaced by the nameplate)	0 ... 1500 bar g

TABLE ROW	VALUES
sizes[1]	V = 10 L
sizes[2]	V = 20 L
sizes[3]	V = 32 L
sizes[4]	V = 50 L

3. Assumptions

A1. All gas-law pressures are absolute. A gauge entry is converted with the atmospheric pressure entered here, so the same numbers read as gauge or as absolute give different volumes. ($p \cdot V^n$ = const on a fixed mass of gas.)
A2. The exponent n is the user's: 1 for a slow discharge with heat exchange (isothermal), towards 1.4 for a fast one (adiabatic). With the same exponent both ways the larger n gives the larger volume for a given ΔV. (Gas-law exponents, closed form.)
A3. The gas process is stated on the sheet. A held accumulator (an emergency stroke, by default) is charged slowly, so isothermally with charge exponent 1 ($V2 = V0 \cdot p0/p2$), and discharged at the exponent n; sizing it with one exponent from the pre-charge to both pressures understates the volume. The one-exponent form (charge exponent = n) is for a gas charged as fast as it is discharged, as in a pulsation or a fast cycle; choose it in the energy-storage duty for that case. The shock and pulsation duty always uses one exponent. (Gas-law sizing of a held accumulator: slow charge, fast discharge.)
A4. The gas is ideal unless a real-gas factor is entered (at least 1: nitrogen's compressibility factor exceeds 1 at high pressure, so the real gas needs more volume, never less); above 200 bar the sheet warns that nitrogen departs from the ideal-gas law and a factor from a gas-property source should be entered. (Real-gas correction as a user factor.)
A5. The pre-charge p0 is defined at the maximum operating temperature T_{max} and V0 is sized there; at the lowest operating temperature T_{min} the same gas delivers only the fraction T_{min}/T_{max} (absolute) of ΔV, so V0 is multiplied by T_{max}/T_{min} for a held accumulator (isothermal charge to p2) and by $(T_{max}/T_{min})^{(1/n)}$ when one exponent n holds both ways (the pre-charge at T_{min} is $p0 \cdot T_{min}/T_{max}$), and the p2/p0 rule is checked with p0 at T_{min}. The temperature range is the user's; with $T_{min} = T_{max}$ the correction is 1. (Manufacturer sizing practice; the gas law at constant mass.)
A6. The pre-charge rules ($p0 \le 0.9 \cdot p1$ and $p2/p0 \le 4$ by default) are records for the accumulator type and make and are checked as PASS/FAIL on ABSOLUTE pressures, whichever basis the pressures are entered on; the right values, and whether the manufacturer states the ratio on gauge or absolute pressures, come from the manufacturer (the same numbers read as gauge pressures give a different ratio). (Rules are records, not built-in limits.)
A7. The bar·litre product is PS times V of the vessel itself: PS is the vessel's maximum allowable pressure in bar gauge (the nameplate or relief setting, never below the highest working pressure p2) and V its internal volume (the selected size unless entered). Without PS and a volume the indication is not available. It indicates whether the 250 bar·L exception of PSSR Schedule 1 can apply; it is not a compliance statement, and PED / UK PER, PSSR examination and the vessel's own design code stay with the manufacturer and the competent person. (Regulatory gating: no pressure-vessel code check.)

A8. mode (Duty) is taken as Energy storage. (Slow clamp-and-hold cycle: stored-energy duty.)
A9. p_{basis} (Pressures entered as) is taken as Absolute. (Pressures entered as absolute bar, as the source numbers are quoted.)
A10. p_{atm} (Atmospheric pressure (gauge to absolute)) is taken as 1.01325 bar a. (Standard atmosphere, derived constant.)
A11. n (Gas exponent n of the discharge (1 isothermal, slow; 1.4 adiabatic, fast); with the same exponent both ways it is also the charge exponent) is taken as 1. (Isothermal: the clamp holds for minutes between cycles.)
A12. f_{rg} (Real-gas factor on V0 (1 = ideal gas; at least 1). It is the ratio of the real to the ideal gas volume for THIS pressure window, not the compressibility factor Z at p_2: from a gas-property source, $f_{rg} = Z_0 \cdot (1/p_1 - 1/p_2)/(Z_1/p_1 - Z_2/p_2)$ for an isothermal discharge, Z at p_0, p_1 and p_2, which is larger than Z_2 for a narrow window) is taken as 1. (Ideal gas; the highest pressure is just above 200 bar, so the sheet warns.)
A13. T_{min} (Minimum operating temperature (the gas is coldest and delivers least here; V0 is scaled by T_{max}/T_{min})) is taken as 30 °C. (Oil temperature range assumed 30 to 50 °C in service (invented for this example); the gas is coldest at 30 °C.)
A14. T_c (Charging temperature) is taken as 20 °C. (Charged in the workshop at 20 °C.)
A15. Energy storage duty: ΔV is the liquid volume the accumulator delivers between p_2 and p_1 over a cycle; n = 1 for a slow discharge, towards 1.4 for a fast one. (Gas-law sizing for a stored-energy duty.)

4. Working

[15] $pa_0 = p_0$
= 90 bar a
= 90 bar a
Pre-charge pressure, absolute

[16] $pa_1 = p_1$
= 100 bar a
= 100 bar a
Lowest working pressure, absolute

[17] $pa_2 = p_2$
= 210 bar a
= 210 bar a
Highest working pressure, absolute

[18] $p2_g = \max(pa_2 - p_{atm}, 0\ bar)$
= max(210 bar a − 1.0132 bar a, 0 bar)
= 208.99 bar g
Highest working pressure, gauge

[21] $V0_{ideal} = dV / [(pa_0/pa_1)^{(1/n)} - (pa_0/pa_2)^{(1/n)}]$ [4](const)
= 14 L / [(0.9) − (0.42857)]
= 29.697 L
Gas volume for the displacement, ideal gas

[22] $f_{Tabs} = (T_{max} / 1\ K) / (T_{min} / 1\ K)$
= 50 degC / 1 K / (30 degC / 1 K)
= 1.066
Absolute temperature ratio T_{max}/T_{min}

[23] $f_T = f_{Tabs}^{(1 / n)}$
= 1.066^(1 / 1)
= 1.066
Temperature scaling of the gas volume: the temperature ratio to the power 1/n (one exponent both ways)

[24] $V0 = V0_{ideal} * f_{rg} * f_T$
= 29.697 L × 1 × 1.066
= 31.656 L
Pre-charge gas volume V0 (accumulator size required)

[25] T_{ratio} = (T_c / 1 K) / (T_{max} / 1 K)
= 20 degC / 1 K / (50 degC / 1 K)
= 0.90716
Charging over maximum absolute temperature
[26] $p0$ = pa_0 * T_{ratio}
= 90 bar a × 0.90716
= 81.645 bar a
Absolute pre-charge pressure at the charging temperature
[27] $p0_{min}$ = pa_0 / f_{Tabs}
= 90 bar a / 1.066
= 84.43 bar a
Absolute pre-charge pressure at the minimum operating temperature
[28] $p0_{set}$ = $p0$
= 81.645 bar a
= 81.645 bar a
Pre-charge to set at the charging temperature (absolute)
[29] u_{r0} = pa_0 / (r_0 * pa_1)
= 90 bar a / (0.9 × 100 bar a)
= 1
Pre-charge against the p0/p1 rule
[30] u_{rp} = (pa_2 / $p0_{min}$) / r_p
= 210 bar a / 84.43 bar a / 4
= 0.62182
Pressure ratio p2/p0 (p0 at the minimum temperature) against the rule
[31] V_{sel} = smallest listed size not below V0
= V0 = 31.656 L, 4 sizes listed
= 32 L
Selected accumulator size
[32] u_V = V0 / V_{sel}
= 31.656 L / 32 L
= 0.98926
Required volume over the selected size
[33] bar_{litres} = PS * V_{sel}
= 230 bar g × 32 L
= 7360 bar·L
Pressure-volume product PS·V (PS as entered, V the selected size)
[34] BL_{pssr} = 250 bar*L [5](1)
= 250 bar·L
PSSR Schedule 1 exception threshold (indication)

5. Results

SYMBOL	VALUE	DESCRIPTION
V0	31.66 L	Accumulator gas volume required
p0	81.64 bar a	Pre-charge, absolute, at the charging temperature
bar_{litres}	7360 bar·L	Pressure-volume product (bar·litres)
V_{sel}	32 L	Selected size
$p0_{set}$	81.64 bar a	Pre-charge to set at the charging temperature
$p0_{min}$	84.43 bar a	Pre-charge, absolute, at the minimum operating temperature
f_T	1.066	Gas volume scaling T_{max}/T_{min}

SYMBOL	VALUE	DESCRIPTION
pssr	"7360 bar·L is at or above 250 bar·L: the exception does not apply; PSSR and PED / UK PER duties follow."	PSSR Schedule 1 250 bar·L exception (indication only)
process	"charged as fast as it is discharged (one exponent n = 1 from the pre-charge to both working pressures)"	Gas process the volume is sized for
u_V	0.99	Selected size utilisation
u_{r0}	1.00	Pre-charge rule utilisation (p0/p1) (governing)
u_{rp}	0.62	Pressure ratio utilisation (p2/p0 at the minimum temperature)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Pre-charge within the p0/p1 rule	u_{r0} = 1.00	u_{r0} ≤ 1	1.00	[6](record)
PASS	Pressure ratio within the p2/p0 rule	u_{rp} = 0.62	u_{rp} ≤ 1	0.62	[6](record)

7. Applicability

WARNING records: tabulated values not taken from a reference record are r_0, r_p (Pre-charge rules); sizes (Accumulator size list). Enter each as one so its source, edition and licence travel with it. [record]
WARNING p_2: the highest working pressure is 210 bar absolute, above 200 bar; nitrogen departs from the ideal-gas law there and the volume shown is the ideal-gas value, too small (reference-EOS nitrogen at 20 °C needs 1.7 to 2.0 times the ideal volume from 450 to 700 bar). Enter a real-gas factor f_{rg} (up to 3) from a gas-property source. [realgas]

Defaults used (inputs left empty):
process: not entered, so the default of 'By duty' is used.

8. Verification

```
Template fp.accumulator@1.5.1 Content hash ebe37e59c3db34a4 Engine calc-engine/1.0.0
Known-answer cases passed 10/10 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. Pressure Systems Safety Regulations 2000 (SI 2000/128).
2. Pressure Equipment Directive 2014/68/EU and UK PER 2016.
3. Esposito, Fluid Power with Applications.
4. p·V^n =.
5. PSSR Schedule.
6. accumulator.rules.